

Seminar 3 - Warm-up Geometry

$K(0, B)$

2-dimensional cases: $B = (i, j)$

3-dimensional cases: $B = (i, j, k)$

(3.1) Det. Cartesian equation for line l :

a) l contains the point $A(-2, 3)$, $\pm 60^\circ$ with Ox .

$\rightarrow$ there are two unit vectors forming a 60° angle with i .

$$u\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), w\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$$

$$\Rightarrow l: \frac{x+2}{1} = \frac{y-3}{\sqrt{3}} \text{ or } l: \frac{x+2}{1} = \frac{y-3}{-\sqrt{3}}$$

or, use: $l: y - y_A = \tan \alpha (x - x_A)$

$$y - 3 = \pm \sqrt{3} (x + 2)$$

$$b) \boxed{l: m_x (x - x_0) + m_y (y - y_0) = 0}$$

$$\text{so } l: 4(x - 1) + 3(y - 7) = 0$$

$$\Rightarrow l: 4x + 3y - 25 = 0$$

$\leftarrow l$ contains the point $B(1, 7)$ and is orthogonal to $m(4, 3)$.

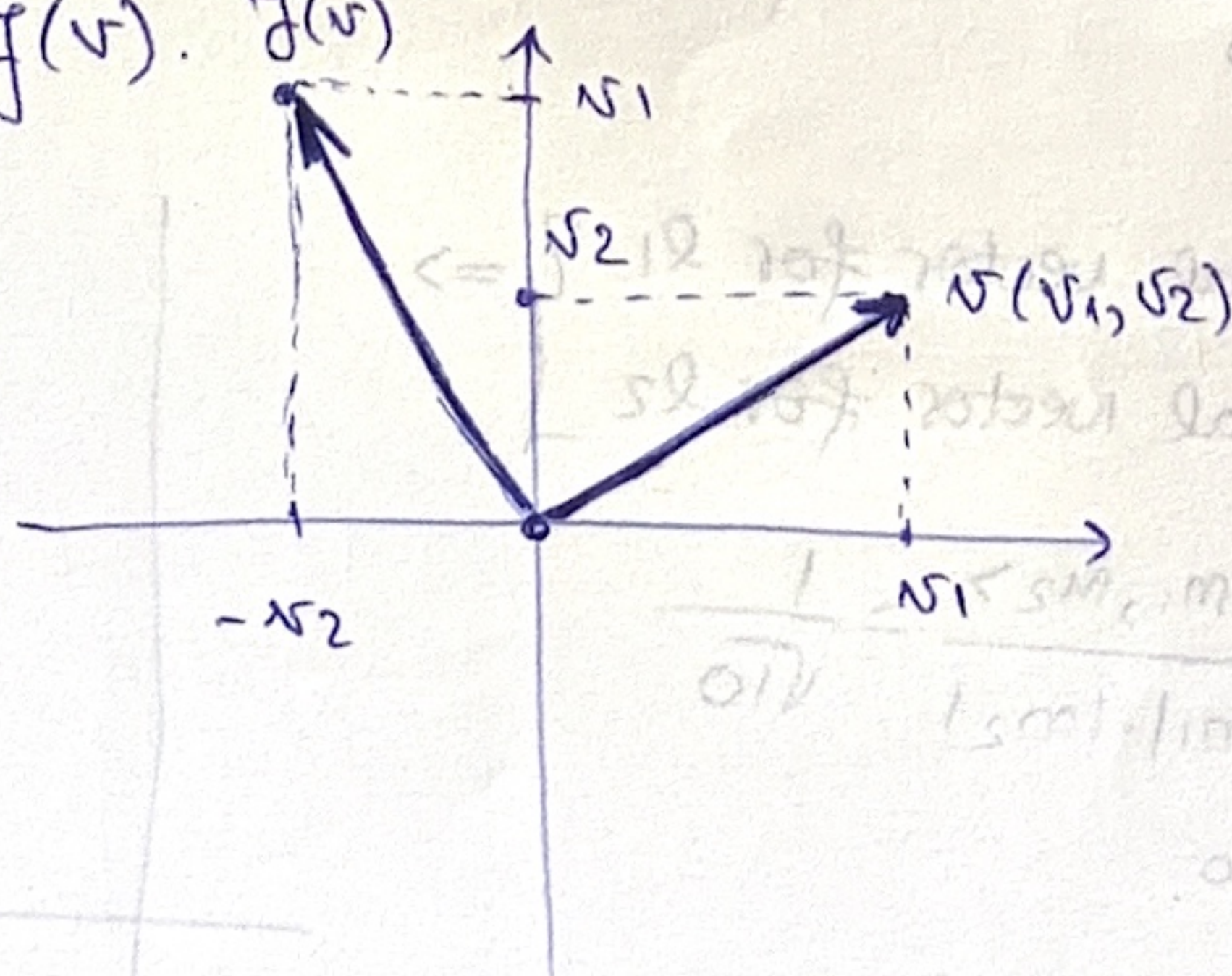
(3.2) Line l in the plane.

a) if $N(v_1, v_2)$ is a direction vector for l then $m(v_2, -v_1)$ is a normal vector for l .

In dimension 2, any vector orthogonal to a direction vector is a normal vector.

$$\langle N, m \rangle = N_1 \cdot m_2 - N_2 \cdot m_1 = 0 \Rightarrow m \perp N \Rightarrow m \text{ is a normal vector.}$$

$$\text{here } m = -J(v).$$



b) if $m(m_1, m_2)$ is a normal vector for l then $v(v_2, -m_1)$ is a direction vector for l .

In dimension 2, any vector orthogonal to a normal vector is a direction vector.

$$\langle m, v \rangle = m_1 m_2 - m_2 m_1 = 0 \Rightarrow v \perp m.$$

$\Rightarrow v$ is a direction vector.

$$\text{here } v = -j(m).$$

3.3 Let $A(1,3)$, $B(-4,3)$, $C(2,9)$ vertices of a triangle. Det:

a) length of the altitude from A

b) line containing the altitude from A.

$$a) |AH| = d(A, BC) = \frac{|1-3+4|}{\sqrt{1+1}} = \frac{5}{\sqrt{2}}$$

$$\downarrow$$

$$\vec{BC}(6,6)$$

$$\downarrow$$

$$BC: \frac{x+4}{1} = \frac{y-3}{1} \Rightarrow x-y+7=0$$

b) $(1,1) \parallel \vec{BC}$, $\vec{BC} \perp \vec{AH} \Rightarrow (1,1)$ is a normal vector for AH.

$$\Rightarrow AH: 1 \cdot (x-1) + 1 \cdot (y-3) = 0$$

y) Det. the angles between $l_1: y=2x+1$
 $l_2: y=-x+2$

Sol 1 $\tan \alpha = 2$, $\tan \beta = -1 < 0$.

$\Rightarrow$ one angle is $\alpha - \beta$

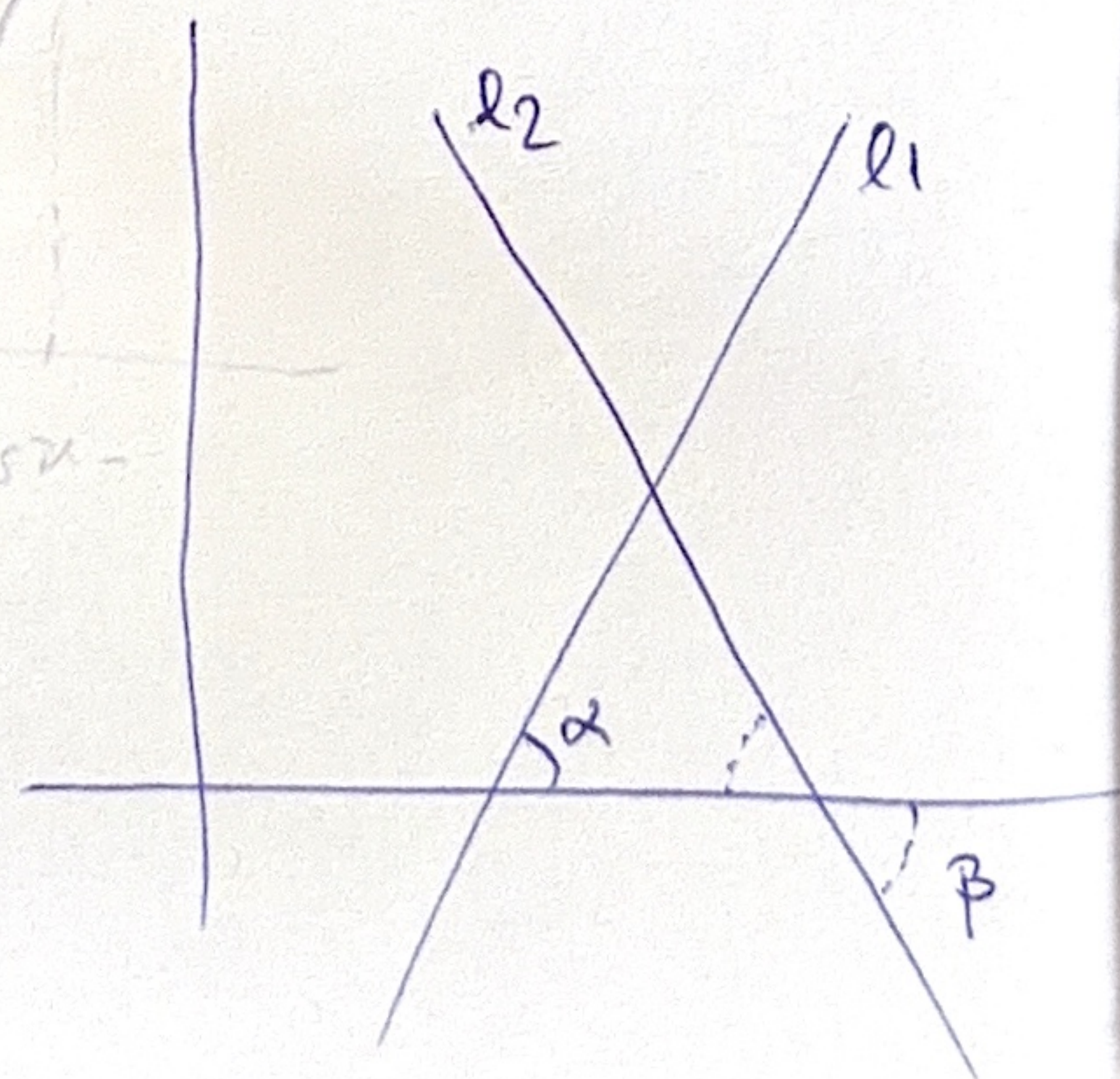
$$\tan(\alpha + \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \cdot \tan \beta} = \frac{2 - (-1)}{1 + 2 \cdot (-1)} = -3$$

$$\Rightarrow \alpha + \beta \approx 71.57^\circ$$

Sol 2 $m_1(2, -1)$ normal vector for l_1 $\Rightarrow$
 $m_2(1, 1)$ normal vector for l_2

$$\Rightarrow \cos \angle(m_1, m_2) = \frac{\langle m_1, m_2 \rangle}{|m_1| \cdot |m_2|} = \frac{1}{\sqrt{10}}$$

$$\Rightarrow \angle(m_1, m_2) \approx 71.57^\circ$$



3.5. Det. Cartesian eq. for the lines situated at distance 4

from the line $12x - 5y - 15 = 0$.

For a point $P(x_0, y_0)$ we have $4 = d(P, l) = \frac{|12x_0 - 5y_0 - 15|}{\sqrt{144 + 25}} \Rightarrow$

$\Rightarrow$ the two lines are:

$$l_{1,2}: 12x - 5y = 15 \pm 52$$

3.6. $A(3, -1)$, $B(9, 1)$, $C(-5, 5)$.

For each pair, det. the line which is equidistant from them.

Sol 1. $P(x_0, y_0)$ we have: $d(A, P) = d(B, P)$

$$\Rightarrow d(A, P)^2 = d(B, P)^2$$

$$\Rightarrow (3 - x_0)^2 + (-1 - y_0)^2 = (9 - x_0)^2 + (1 - y_0)^2$$

$$\Rightarrow 9 - 6x_0 + x_0^2 + 1 + 2y_0 + y_0^2 = 81 - 18x_0 + x_0^2 + 1 - 2y_0 + y_0^2$$

$$\Rightarrow 12x_0 + 4y_0 - 72 = 0$$

$$\text{So } l: 3x_0 + y_0 - 18 = 0$$

Sol 2. $\vec{AB}(6, 2)$ is a normal vector for l .

$M(6, 0)$ is the midpoint of $[AB] \Rightarrow M \in l$.

$$\Rightarrow l: 6(x - 6) + 2y = 0$$

$$\Rightarrow l: 3x + y - 18 = 0$$

3.7. Consider a basis consisting of vectors $v_1(0, 1)$ and $v_2(2, 1)$. Use

the Gram-Schmidt process to find an orthonormal basis containing

v_1 .

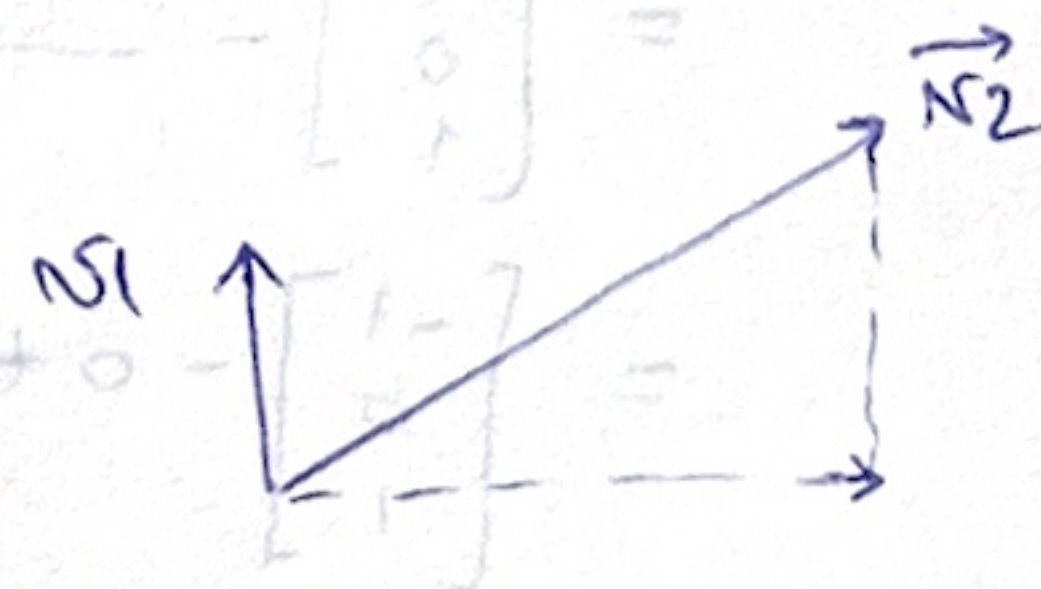
We keep v_1 unchanged and correct v_2 .

$$v_1' = v_1$$

$$v_2' = v_2 - \text{Pr}_{v_1}(v_2) = v_2 - \frac{\langle v_1, v_2 \rangle}{\langle v_1, v_1 \rangle} \cdot v_1$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \frac{1}{1} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$



my
Hello
Hello
or

Then, we make sure that they have unit length.

$$N_1'' = \frac{N_1}{|N_1|} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$N_2'' = \frac{N_2}{|N_2|} = \frac{1}{2} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$\Rightarrow$ the orthogonal basis is: $\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$

3.8) $a(2,1,0)$, $b(0,-2,1)$. Det. the angles between the diagonals of a parallelogram spanned by a and b .

$$d_1 = a + b = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$d_2 = a - b = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

$$\cos \angle(d_1, d_2) = \frac{\langle d_1, d_2 \rangle}{|d_1| \cdot |d_2|} = \frac{4 - 3 - 1}{\sqrt{2} \cdot \sqrt{5}} = 0$$

$$\Rightarrow \angle(d_1, d_2) = 90^\circ$$

3.9) $N_1(0,1,0)$, $N_2(2,1,0)$, $N_3(-1,0,1)$.

Use the Gram-Schmidt process to find an orthonormal basis containing N_1 .

$$W_1 = N_1$$

$$W_2 = N_2 - \frac{\langle W_1, N_2 \rangle}{\langle W_1, W_1 \rangle} \cdot W_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \frac{\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \rangle}{\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \rangle} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$W_3 = N_3 - \frac{\langle W_1, N_3 \rangle}{\langle W_1, W_1 \rangle} \cdot W_1 - \frac{\langle W_2, N_3 \rangle}{\langle W_2, W_2 \rangle} \cdot W_2 =$$

$$= \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \frac{\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \rangle}{1} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \frac{\langle \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \rangle}{4} \cdot \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - 0 + \frac{1}{2} \cdot \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\Rightarrow W_1, W_2, W_3$ is an orthogonal basis.

$\Rightarrow$ an orthogonal basis is $\frac{W_1}{|W_1|} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$\frac{W_2}{|W_2|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\frac{W_3}{|W_3|} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

3.10. Show that a parallelepiped with faces in the planes $2x + y - 2z + 6 = 0$, $2x - 2y + z - 8 = 0$, $x + 2y + 2z + 1 = 0$ is rectangular.

• A parallelepiped is rectangular if its faces are orthogonal to each other ($\Rightarrow$) the planes containing the faces have 90° dihedral angles.

• normal vectors of the given planes are: $m_1(2, 1, -2)$ and $m_2(2, -2, 1)$
 $m_3(1, 2, 2)$.

one checks that $\langle m_1, m_2 \rangle = 0$
 $\langle m_1, m_3 \rangle = 0$
 $\langle m_2, m_3 \rangle = 0$

3.11. The vertices of the tetrahedron are $A(-1, -3, 1)$, $B(5, 3, 8)$, $C(-1, -3, 5)$ and $D(2, 1, -4)$. Height of the tetrahedron relative to the face ABC?

$d(D, ABC)$

$$\vec{AB} = \begin{bmatrix} 6 \\ 6 \\ 7 \end{bmatrix}$$

$$\vec{AC} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$$

$$\Rightarrow ABC: \begin{vmatrix} x+1 & y+3 & z-1 \\ 6 & 6 & 7 \\ 0 & 0 & 4 \end{vmatrix} = 0$$

$$\Rightarrow 24(x+1) - 24(y+3) = 0$$

$$\Rightarrow ABC: x - y - 2 = 0$$

$$\Rightarrow d(D, ABC) = \frac{|2 - 1 - 2|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$